

Name: _____

Assume I am given a short (2-byte integer) virtual address. Assume the first 2 bits represent the segment, while the remaining bits represent the offset within the segment.

Name	Address	Base	Bounds
Code	00	16 000	1 000
Heap	01	32 000	16 000
Stack	10	62 000	12 000

For each given address, state the segment in which the address is located. State whether the address is within the bounds, and if so, state the corresponding physical address.

1. Virtual Address : 128, Binary Representation : 00000000 10000000
2. Virtual Address : 17125, Binary Representation : 01000010 11100101
Hint : 1011100101 equals 741.
3. Virtual Address : 34789, Binary Representation : 10000111 11100101
Hint 11111100101 equals 2021.